Inorganic Chemistry

Article

Cite This: Inorg. Chem. 2017, 56, 13078-13085

pubs.acs.org/IC

Lead-Free Hybrid Material with an Exceptional Dielectric Phase Transition Induced by a Chair-to-Boat Conformation Change of the Organic Cation

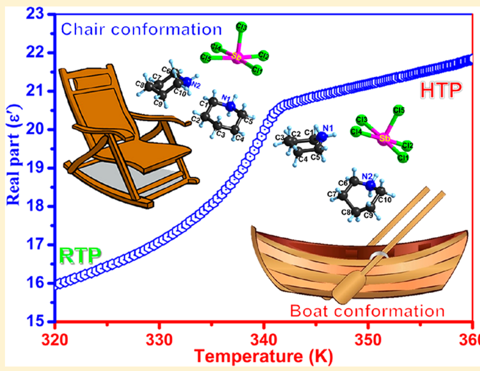
Shiguo Han,[†,‡] Jing Zhang,[†] Zhihua Sun,*[,†] Chengmin Ji,[†] Weichuan Zhang,[†] Yuyin Wang,[†] Kewen Tao,[†] Bing Teng,[‡] and Junhua Luo*[,†]

[†]State Key Laboratory of Structure Chemistry, Fujian Institute of Research on the Structure of Matter, Chinese Academy of Sciences, Fuzhou 350002, P. R. China

[‡]College of Physics, Qingdao University, Qingdao 266071, P. R. China

S *Supporting Information*

ABSTRACT: Hybrid organic−inorganic perovskite materials have demonstrated great potential in the field of photovoltaics and photoelectronics. On the basis of the high degree of structural flexibility and compatibility, diverse molecular functional materials have been assembled by modifying the length of the organic components and/or dimensionality of the inorganic frameworks. In this paper, we present a chiral lead-free organic−inorganic hybrid, (piperidinium)$_2$SbCl$_5$ (**1**), which follows the one-dimensional inorganic frameworks of the corner-sharing SbCl$_6$ octahedra. Strikingly, **1** displays a dielectric phase transition at T_c = 338 K, changing from the chiral space group of $P2_12_12_1$ to polar $Pna2_1$ upon heating. Crystal structure analyses reveal that an unusual thermally activated conformation change of the piperidinium cations affords the driving force to the phase transition of **1**. That is, organic piperidinium moieties display a chairlike conformation below T_c, which transforms to a boatlike structure above T_c. Such an unprecedented change is strongly coupled to the dielectric transition along with notable steplike anomalies, which suggest that **1** could be used as a potential switchable dielectric material. Besides, the temperature-dependent conductivity and theoretical analysis of its electronic structure disclose the semiconducting behavior of **1**. This study paves the pathway to the design of new lead-free semiconducting perovskites with targeted properties for optoelectronic application.



■ INTRODUCTION

Hybrid organic−inorganic perovskite materials (HOIPMs) have received extraordinary research attention in the past decade because of their fascinating potential in optoelectronic applications.[1−7] Such hybrids combine the excellent merits of organic and inorganic components, allowing for a high degree of structural tunability and rich physical properties.[8,9] Usually, the inorganic frameworks offer opportunities for semiconducting behavior and a wide range of band gaps.[10−12] In contrast, organic counterparts provide the possibility of structural diversities and dynamic behavior, such as the random motions of molecular dipoles.[13−15] As a consequence, the flexible organic moieties afford large freedom through low barriers to reorientation within the lattices, which greatly contribute to dielectric/ferroelectric responses.[16−24] As demonstrated by the recently reported two-dimensional hybrid perovskites, such as (benzylammonium)$_2$PbCl$_4$ and (cyclohexylammonium)$_2$PbBr$_{4-4x}$I$_{4x}$, the dramatic order−disorder changes of organic components induce their structural phase transitions, along with giant dielectric responses and ferroelectric properties.[25,26] In this context, the delicate introduction of organic cationic moieties with dynamic reorientations or motions has been conceived as an effective strategy to construct new electric-ordered materials.

Methylammonium lead halides (i.e., CH$_3$NH$_3$PbX$_3$, with X = Cl, Br, and I) are the most notable hybrid perovskites and have made a breakthrough of power conversion efficiencies for photovoltaic solar cells.[27−32] Remarkable semiconducting properties have been identified in crystals of CH$_3$NH$_3$PbI$_3$, including large absorption coefficients, outstanding ambipolar charge transport, pimping exciton binding energy and high tolerance to defects, etc.[33−37] In addition, the phase transition and potential ferroelectricity were also observed in CH$_3$NH$_3$PbI$_3$, which originated from the dynamic disordering of CH$_3$NH$_3^+$ cations.[38−40] For instance, a large polarization of ~38 μC/cm^2 has been theoretically estimated in CH$_3$NH$_3$PbI$_3$, and the ferroelectric domain walls were also observed from the solution-processed thin films.[41−47] It was deemed that the polarization also makes a contribution to their photoelectric behavior.[48] However, all of these hybrid perovskites contain poisonous metal lead, which will aggravate the pollution

Received: July 25, 2017
Published: October 25, 2017

DOI: 10.1021/acs.inorgchem.7b01863
Inorg. Chem. 2017, 56, 13078−13085

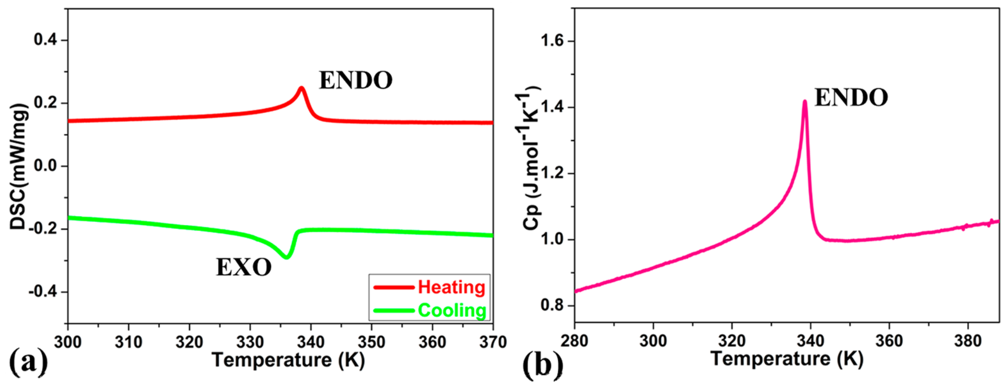


Figure 1. Temperature-dependent heat-flow measurements: (a) DSC; (b) specific heat measurements of **1**.

impacts of the environment and become an obstacle for their further commercialization. As the preliminary indicator, the phase transition is inseparable for the exploration of ferroelectric materials.[49−55] Under this circumstance, it is highly desirable to explore new lead-free hybrid phase-transition materials for specified device applications.

We have reported a series of lead-free HOIPMs, such as (*N*-methylpyrrolidinium)$_3Sb_2Br_9$ and (*N*-methylpyrrolidinium)$_3Sb_2Cl_{9-9x}Br_{9x}$.[56,57] These materials display a structural phase transition accompanied by remarkable ferroelectric semiconducting properties. However, all of these phase-transition mechanisms belong to the order−disorder transformation of organic cations. In a subsequent study, we further performed the chemical modification of these hybrid perovskites for phase-transition properties, and we present a chiral organic−inorganic hybrid, (piperidinium)$_2SbCl_5$ (**1**), which consists of a one-dimensional (1D) perovskite-like anionic framework constructed by corner-sharing $SbCl_6$ coordinated octahedra. More strikingly, compound **1** displays a dielectric phase transition at T_c = 338 K, transforming from a chiral space group of $P2_12_12_1$ (below T_c) to a polar one of $Pna2_1$ (above T_c). It is unprecedented that its phase transition is induced by an unusual conformation variation of piperidinium cations, namely, a “chair-to-boat” change of the spatial conformation. According to what we know, this might be the first discovery of a solid-state structural phase transition triggered by concrete conformation changes. The invertible switching of dielectric constants between two distinct states discloses its potential as a switchable dielectric material.[58−60] Because of the chainlike inorganic framework, **1** exhibits notable semiconducting properties that are confirmed by the positive temperature-dependent conductivity and theoretical analysis of the electronic structure. This study opens up new possibilities of developing electric-ordered materials accompanied by excellent semiconducting behavior.[61−63]

EXPERIMENTAL SECTION

Materials and Methods. A single crystal of **1** was synthesized by slow solvent evaporation at room temperature. First, antimony trichloride salts ($SbCl_3$) were prepared by the reaction of antimony trioxide with excessive hydrochloric acid in a cold bath. After stirring for 20 min, piperidine was added in the molar ratio of 2:1 with antimony trioxide to the bath. Then, the reaction mixture was stirred with heating for 20 min to dissolve the components thoroughly. Colorless crystals were easily acquired by means of slow evaporation of the solution. A Vario EL cube elemental analyzer was used to check the elemental contents (C, H, and N). Calcd: C, 25.48; H, 5.13; N, 5.94. Found: C, 25.32; H, 5.03; N, 5.92.

Powder X-ray Diffraction (PXRD). PXRD was used to check the phase purity of **1**, and the result matches well with the simulated data, as seen in Figure S1. In addition, a Mini Flex II powder X-ray diffractometer was used to collect the variable-temperature PXRD data.

Thermal Measurement. A Netzsch DSC 200 F3 calorimeter was used to record differential scanning calorimetry (DSC) and specific heat (C_p) data. Also, a Netzsch STA 449C unit was used to collect thermogravimetric and differential thermal analysis (TG/DTA) data.

Dielectric Constant Measurement. Dielectric constant tests were performed on the pressed-powder pellets that were covered by silver conducting glue. A Tonghui TH2828A analyzer was used to record the temperature variability of the real part (ε') of **1** with a frequency of 700 kHz.

Single-Crystal X-ray Diffraction. A Super-Nova diffractometer with Mo Kα radiation (λ = 0.71073 Å) was used to measure the structure data at room temperature (300 K) and high temperature (353 K). Crystallographic data in CIF format are given in the following: CCDC 1541036 for 300 K and 1541037 for 353 K. The *Crystalclear* software package (Rigaku, 2005) was utilized to collect structure information. The *SHELXLTL* software package was used to solve and refined variable-temperature crystal structures.[64] Crystallographic data are listed in Table S1.

Circular Dichroism (CD) Spectroscopy. Solid-state CD spectroscopy of compound **1** was performed on a MOS-450 spectrometer within a range of 200−500 nm at 295 K using KBr pellets.

Second-Harmonic-Generation (SHG) Measurement. A fundamental laser (Vibrant 355 II, OPOTEK) beam with a low divergence was used to detect the SHG properties of **1** at 295 K.

Ultraviolet−Vis (UV−Vis) Spectrometry. UV−vis diffuse-reflectance spectrometry was performed on a PerkinElmer Lambda 950 UV−vis−IR spectrophotometer.

Alternating-Current (ac) Conductivity. The electrical property of ac conductivity (σ_{ac}) was investigated with a direct-current two-terminal method using single crystals of **1**. The temperature dependence of conductivity was carried out in the heating mode.

RESULTS AND DISCUSSION

Thermal Properties. The DSC test is an effective tool to confirm the existence of an invertible thermally induced phase transition.[65−67] Here, compound **1** was subjected to preliminary DSC measurement (Figure 1a), which shows an endothermic peak (338.5 K) in the heating mode and an exothermic peak (336.0 K) in the cooling mode, revealing an invertible phase transition. Besides, from the C_p−T curve, the existence of a phase transition is further confirmed, as seen in Figure 1b. The C_p−T curve shows a sharp peak at 338.5 K, which is in accordance with the DSC result. The enthalpy (ΔH) related to the endothermic peak was estimated to be 2.166 J/g. Therefore, the total entropy change (ΔS) was

DOI: 10.1021/acs.inorgchem.7b01863
Inorg. Chem. 2017, 56, 13078−13085

calculated to be 3.0158 J/mol·K by following the equation $\Delta S = \Delta H/T_c$. The value of N was calculated to be 1.44 in light of the Boltzmann formula $\Delta S = R \ln N$ (N is the ratio of possible configurations and R is the gas constant). To our knowledge, such a value of N implies a more complicated phase transition than the previous order−disorder mechanism (N = 2).[68] For the sake of convenience, we labeled T_c as the boundary of the room-temperature phase (RTP) and high-temperature phase (HTP). Besides DSC and C_p studies, the pronounced thermal anomaly was also observed by TG/DTA, further confirming the phase transitions of **1** (Figure S2).

Crystal Structure Discussion. In order to understand the mechanism of phase transition between RTP and HTP, variable-temperature single-crystal X-ray diffraction was carried out at 300 and 353 K, respectively. The results suggest that the asymmetric unit of **1** is made up of one $[SbCl_5]^{2-}$ anion and two protonated piperidinium cations at both RTP and HTP (Figure 2). The structure of **1** at RTP belongs to the

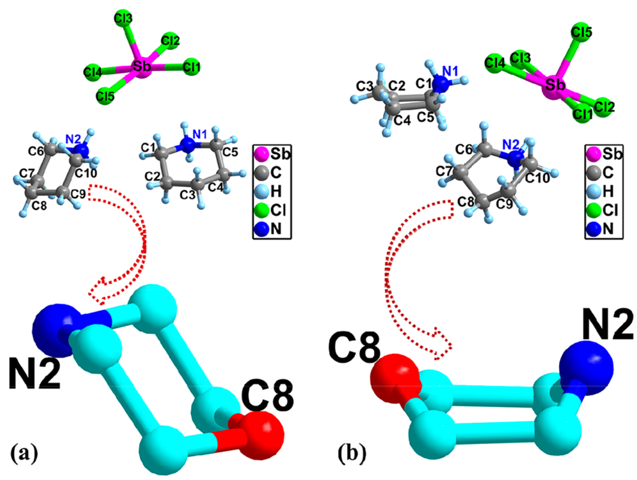


Figure 2. Top: Asymmetric units of **1** (a) at 300 K and (b) at 353 K. Bottom: Conformation transformation of the piperidinium cation.

orthorhombic system with a chiral space group $P2_12_12_1$ and a point group D_2. The cell parameters of **1** are a = 8.4914(6) Å, b = 12.2708(8) Å, c = 18.1989(14) Å, and V = 1896.30(2) Å^3. With temperature increasing to the HTP (353 K), **1** still crystallizes in the orthorhombic system. However, the cell parameters of **1** change to a = 12.3177(3) Å, b = 18.4433(4) Å, c = 8.3976(3) Å, and V = 1907.76(9) Å^3. Accordingly, the space group of **1** transforms from $P2_12_12_1$ to $Pna2_1$, and its point group becomes a polar one, C_{2v}. The remarkable differences of the crystallographic space group between RTP and HTP can be easily comprehended from the microscopic structural changes of **1** during its phase transition. As is vividly depicted in the bottom of Figure 2, the piperidinium cation resembles a "chair" at RTP; however, it becomes to a "boat" when the temperature rises to HTP. In particular, for the piperidinium cation of **1** at RTP, the atoms of N1 (or N2) and C3 (or C8) are situated below and above the coplane, respectively, behaving as a "chair". In contrast, at HTP, the atoms move to the same side of the coplane and the structure of the piperidinium cation changes to a "boatlike" configuration. That is, **1** undergoes a chair-to-boat conformation transformation during its phase transition, which differs from the previous order−disorder mechanism. To our knowledge, this should be the first report on the solid-state structural phase transition triggered by concrete conformation changes.

As for the packing structure, **1** is featured by a 1D chainlike inorganic framework, which is closely linked by piperidinium cations at both RTP and HTP, as shown in Figure 3. In detail,

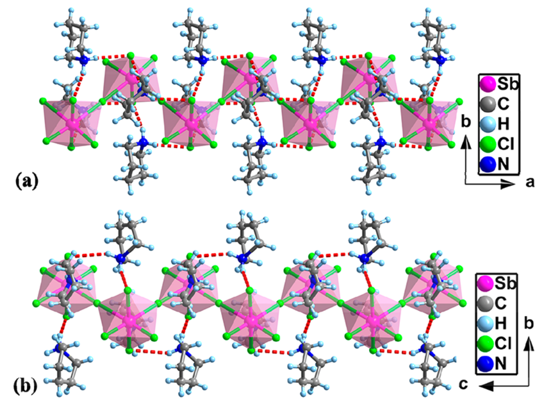


Figure 3. 1D structure of **1** (a) at 300 K along the c axis and (b) at 353 K along the a axis. Red imaginary lines represent the N−H···Cl hydrogen bonds.

the organic piperidinium cations are bonded to an inorganic anionic framework through N−H··Cl hydrogen bonds (Tables S2 and S3). It is known that the inorganic frameworks of HOIPS offer opportunities for semiconducting behavior and a wide range of band gaps. For **1**, the 1D inorganic $[SbCl_5]^{2-}$ chain would contribute to its charge-transport properties, just like some tin- and lead-based perovskites. Moreover, the trivalent antimony ion is isoelectronic with Sn^{2+} and Pb^{2+}, and the electronegativities and ionic radii of these elements are similar. Thus, compound **1** is expected to exhibit latent semiconducting behavior.

Although piperidinium cations of **1** undergo exceptional chair-to-boat conformation transformation during its phase transition, the inorganic framework only exhibits tiny changes. As depicted in Figure 4, the central antimony atoms of $SbCl_6$

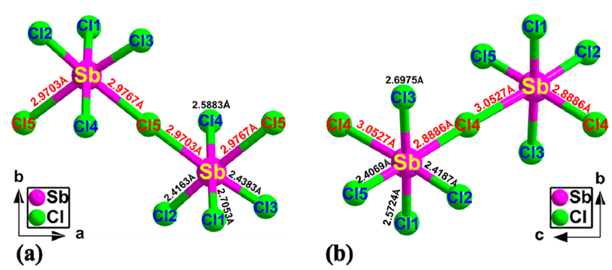


Figure 4. Corner-sharing $SbCl_6$ octahedra of the 1D inorganic framework of **1** at 300 K (a) and 353 K (b).

octahedra are coordinated by six bromide atoms, one of which behaves as the bridging linker. As a result, two $SbCl_6$ octahedra share one bridging corner (Cl5 at RTP or Cl4 at HTP), and the corner-sharing $SbCl_6$ octahedra construct a 1D perovskite-like inorganic framework. In addition, it is obvious that the geometry of this structural octahedron is slightly distorted, as deduced from the difference between the Sb−Cl bond lengths. Besides, the coordinated distortion can also be confirmed from the different bond angles (Cl−Sb−Cl) of the $SbCl_6$ irregular octahedra (Table S4).

DOI: 10.1021/acs.inorgchem.7b01863
Inorg. Chem. 2017, 56, 13078−13085

Variable-Temperature PXRD (VT-PXRD). In general, VT-PXRD offers the most powerful information on phase-transition materials. As depicted in Figure 5, the PXRD data

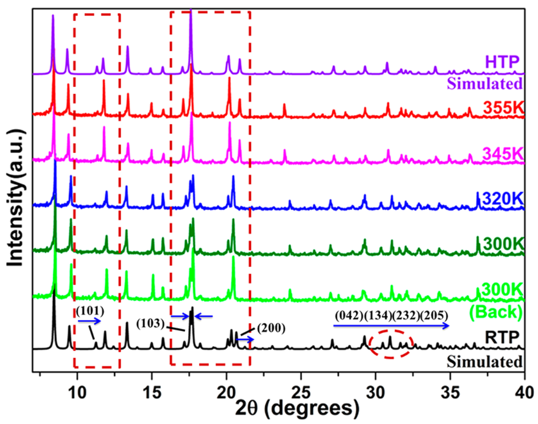


Figure 5. VT-PXRD patterns of **1**. The Miller indexes were calculated from the single-crystal structure.

collected at RTP (300 and 320 K) and HTP (345 and 355 K) are in well accordance with relevant simulated data (the black and purple lines). However, drastic changes were found between the RTP and HTP data, strongly ensuring the existence of a phase transition in **1**. For instance, as the temperature increases above T_c, the diffraction peaks at 11.46°, 20.94°, 30.83°, 31.23°, 31.94°, and 32.34°, relevant to the Bragg diffraction patterns of the (010), (200), (042), (134), (232), and (205) planes at RTP, exhibit obvious shifts toward the high-angle direction. In addition, the two peaks at 18.00° merge into one peak as the temperature moves from RTP to HTP. Generally, the decrease in the number of diffraction peaks indicates enhancement of the crystal symmetry. The symmetry transformation of **1** (i.e., from the point group D_2 to C_{2v}) agrees fairly well with this concept. It is obvious that the PXRD data acquired upon cooling back to 300 K from HTP (the light-green line) are in accordance with the patterns collected at 300 K (the dark-green line). Such a consequence solidly ensures the invertible phase transition in **1**.

Solid-State CD Spectra and SHG Characterization. CD spectroscopy is one of the simple and quick ways to obtain structural information on molecular crystals. To examine the chiroptical activities of **1**, solid-state CD spectra were investigated at 295 K. The CD spectra of compound **1** display a strong Cotton effect around 230 nm (Figure S3), solidly confirming that **1** belongs to a chirality structure at room temperature.[69,70] This finding is well in accordance with the single-crystal X-ray diffraction consequences, which disclose that the space group of **1** is $P2_12_12_1$ at RTP. Because compound **1** has a chiral space group of $P2_12_12_1$ at room temperature, which belongs to the noncentrosymmetric structure, it is expected to exhibit interesting a second-order nonlinear-optical (NLO) response. Here, we further record the SHG signal of **1** using the fundamental laser beam at 295 K. As depicted in Figure S4, **1** shows an obvious frequency-doubled signal around 532 nm, which is well consistent with its noncentrosymmetric structure. Specifically speaking, as the standard sample, KDP was used for comparison. The SHG intensities of **1** were 0.2 times those of KDP, indicating that **1** might be used as a possible hybrid organic−inorganic NLO material.

Dielectric Properties. The changes of the dielectric constants between RTP and HTP represent the degree of electric polarizability of the materials. So, the temperature-dependent dielectric constant measurement can be seen as an indicator for phase transition.[71] Structure analyses have disclosed the conformation transformation of **1** during its phase transition. To further confirm this feature, we measured its temperature-dependent complex dielectric permittivities ($\varepsilon = \varepsilon' - i\varepsilon''$) in the heating/cooling cycle on pressed-powder pellets. Figure 6 shows the real part (ε') of **1** at a frequency of

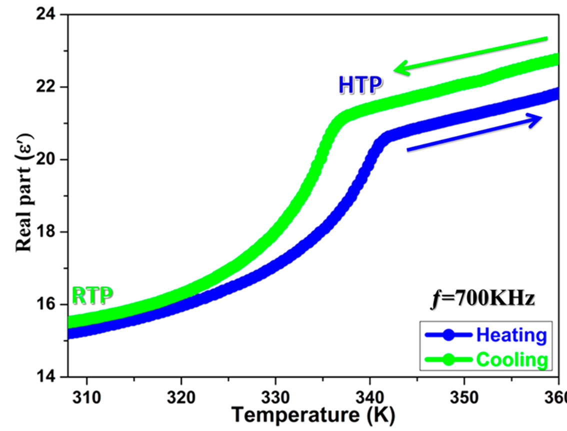


Figure 6. Temperature variable of the real part (ε') of the complex permittivity for **1**.

700 kHz within the range of 300−360 K. Obviously, we can see a couple of steplike dielectric anomalies at 338.5 and 336 K in the heating and cooling procedure, respectively. This finding is well compatible with the DSC result. In detail, upon cooling, the value of ε' declines from 22 to 15. The value of the dielectric constant at the HTP state (f = 700 kHz) is ∼40% larger than that of the RTP state, which discloses that **1** can be used as a "dielectric switch". This behavior, that is, the invertible switching of the dielectric constants between high and low states, indicates the splendid dielectric switchable property of **1**.

Semiconducting Properties. Because the structure of **1** features a 1D inorganic framework, it is expected to display interesting semiconducting properties. Here, we conducted UV−vis diffuse-reflectance spectroscopy to explore the semiconducting properties of **1**. It is common knowledge that the semiconductor band-gap type can be divided into direct and indirect band gaps based on its light absorption features. Here, for convenience, the direct-band-gap semiconductor ZnO was selected for comparison. As shown in Figure 7, like ZnO, the UV−vis absorption spectra of **1** exhibit a sharp edge in the range of the UV spectra, indicating that compound **1** is a direct-band-gap semiconductor. Theoretical analysis shows the semiconducting band gap calculated by the Tauc formula:[72]

$$[h\nu F(R_\infty)]^{1/n} = A(h\nu - E_g)$$

where h is the Planck constant, ν is the vibrational frequency of the oscillator, $F(R_\infty)$ is the Kubelka−Munk function, A is the constant of proportionality, and E_g is the band gap.[73] The value of n is based on its band-gap type (n = 2 for the indirect band gap and n = $^1/_2$ for the direct band gap). Therefore, the band gap E_g of **1** can be calculated from the Tauc plot (inset of Figure 7). The estimated band gap is 3.23 eV for **1**, comparable

DOI: 10.1021/acs.inorgchem.7b01863
Inorg. Chem. 2017, 56, 13078−13085

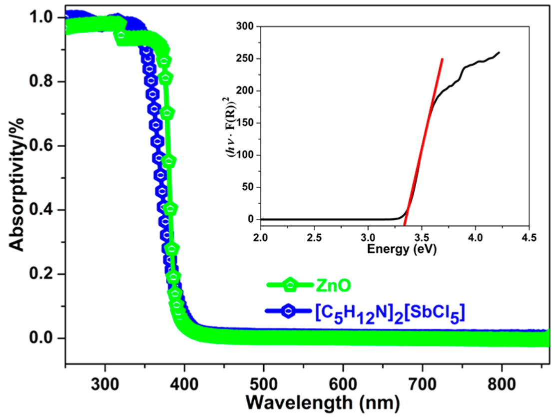


Figure 7. UV−vis absorption spectra for **1** and ZnO. Inset: Calculated band gap of **1**.

to other analogues with **1**, such as $(CH_3NH_3)PbCl_3$ with a band gap of 2.97 eV.[74]

Moreover, we measured the temperature-dependent conductivity of **1** to explore its semiconducting properties. As shown in Figure 8, with increasing temperature, the positive

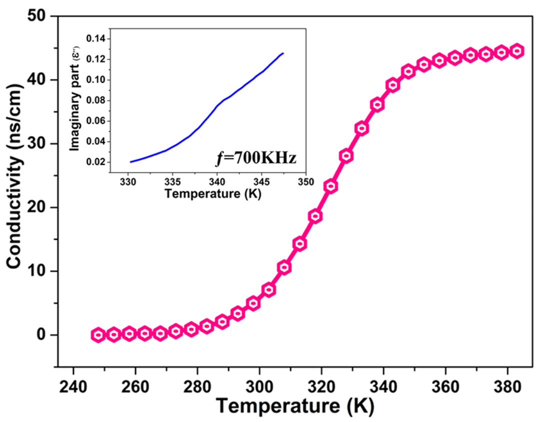


Figure 8. Temperature-dependent conductivity of **1**. Inset: Imaginary part (ε'') of the complex dielectric permittivity of **1**.

slope of the conductivity reveals that **1** is a semiconductor. In addition, the measured temperature-dependent ac conductivity of **1** is consistent with that of the imaginary part (ε'') of the dielectric constant (f = 700 kHz; inset of Figure 8), according to the formula $\sigma_{ac} = \omega\varepsilon''\varepsilon_0$ (ω is the circular frequency and ε_0 is the dielectric constant of free space).[75,76] Therefore, the positive slope of the conductivity with increasing temperature and the sharp rise of the absorptivity indicate that **1** is a semiconductor.[77,78]

Furthermore, in order to explore the semiconducting electronic mechanism of **1**, we calculated the band structure and energy gap, making use of density functional theory. As depicted in Figure 9, the maximum of the valence band and the minimum of the conduction band are located at the X point, indicating that **1** is a direct-band-gap semiconductor. This discovery is in well accordance with the UV−vis absorption spectral result. In addition, the value of the theoretical band gap is 3.38 eV, which coincides with the laboratorial value of 3.23 eV.

Additionally, for the partial density of states of **1**, as depicted in Figure 10, obviously, the organic portions (C 2s, H 1s, and N

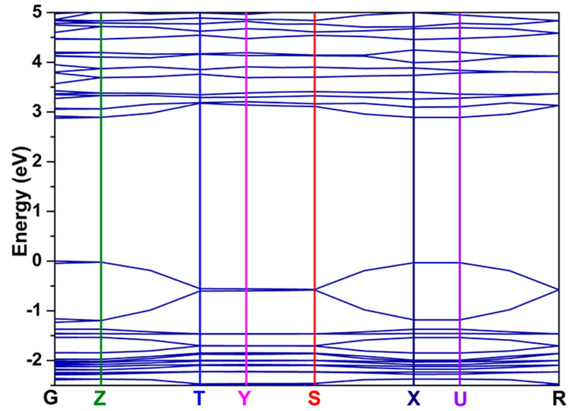


Figure 9. Calculated band structure of **1**.

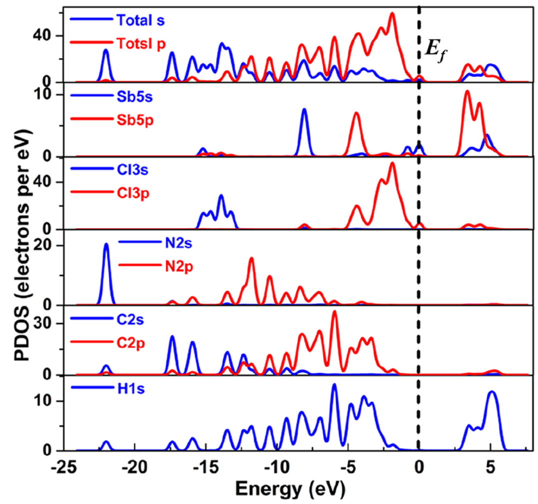


Figure 10. Partial density of states of **1**.

2s2p states) overlap completely within the range from −17 to −1 eV. This result reveals intense covalent forces in the N−H and C−H bonds. Besides, in the neighborhood of the Fermi level (E_f) of the partial density of states, quite strong overlap is observed between Sb s/p and Br s/p. Moreover, for the whole energy region (−16 to −12, −9 to 0, and 2.5−5.5 eV), Sb s/p and Cl s/p overlap distinctly, indicating the intense interactions in the Sb−Cl bonds. The intensive conduction bands mainly result from the states of Sb 5p, while the flat valence bands originate from the Cl 3p states. Such coupling reveals that the band gap of **1** is determined by the electronic structure of the 1D inorganic chain, resembling some lead- and antimony-based perovskite hybrids.[79]

■ CONCLUSION

In summary, we have successfully demonstrated one lead-free hybrid organic−inorganic hybrid, $[C_5H_{12}N]_2[SbCl_5]$, which undergoes a high-temperature phase transition induced by an unusual chair-to-boat conformation change of the piperidinium cations. The dielectric constants can be switched between two distinct states, which discloses its potential as a switchable dielectric material. In addition, with increasing temperature, the positive slope of the conductivity suggests that **1** could be used as a lead-free semiconductor. Theoretical analysis of the electronic structure and energy gap suggests that **1** displays direct-band-gap behavior, and the estimated band gap E_g is 3.38

DOI: 10.1021/acs.inorgchem.7b01863
Inorg. Chem. 2017, 56, 13078−13085

eV. Our study opens up new ways to develop electric-ordered materials accompanied by excellent semiconducting behavior.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.inorgchem.7b01863.

Experimental and calculated PXRD patterns, TG/DTA curves, CD spectra, experimental SHG signal, crystal data and structure refinement, hydrogen bonds, and bond angles (PDF)

Accession Codes

CCDC 1541036−1541037 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

■ AUTHOR INFORMATION

Corresponding Authors

*E-mail: sunzhihua@fjirsm.ac.cn.

*E-mail: jhluo@fjirsm.ac.cn.

ORCID

Junhua Luo: 0000-0002-7673-7979

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the NSFC (Grants 21601188, 21525104, 21571178, 21622108, 91422301, 21373220, 51402296, 51502288, and 51502290), the Strategic Priority Research Program of the Chinese Academy of Sciences (Grant XDB20000000), and the Youth Innovation Promotion of CAS (Grant 2014262). Z.S. acknowledges support from the State Key Laboratory of Luminescence and Applications (Grant SKLA-2016-09).

■ DEDICATION

Dedicated to the 80th birthday for Prof. Academician Xin-Tao Wu from FJIRSM, CAS.

■ REFERENCES

(1) Saparov, B.; Mitzi, D. B. Organic−Inorganic Perovskites: Structural Versatility for Functional Materials Design. *Chem. Rev.* **2016**, *116*, 4558−4596.

(2) Green, M. A.; Ho-Baillie, A.; Snaith, H. J. The emergence of perovskite solar cells. *Nat. Photonics* **2014**, *8*, 506−514.

(3) Møller, C. K. Crystal Structure and Photoconductivity of Cæsium Plumbohalides. *Nature* **1958**, *182*, 1436−1436.

(4) Zhu, H. M.; Fu, Y. P.; Meng, F.; Wu, X. X.; Gong, Z. Z.; Ding, Q.; Gustafsson, M. V.; Trinh, M. T.; Jin, S.; Zhu, X. Y. Lead halide perovskite nanowire lasers with low lasing thresholds and high quality factors. *Nat. Mater.* **2015**, *14*, 636−642.

(5) Gao, P.; Gratzel, M.; Nazeeruddin, M. K. Organohalide lead perovskites for photovoltaic applications. *Energy Environ. Sci.* **2014**, *7*, 2448−2463.

(6) Snaith, H. J. Perovskites: The Emergence of a New Era for Low-Cost, High-Efficiency Solar Cells. *J. Phys. Chem. Lett.* **2013**, *4*, 3623−3630.

(7) You, Y. M.; Liao, W. Q.; Zhao, D. W.; Ye, H. Y.; Zhang, Y.; Zhou, Q. H.; Niu, X. H.; Wang, J. L.; Li, P. F.; Fu, D. W.; Wang, Z. M.; Gao, S.; Yang, K. L.; Liu, J. M.; Li, J. Y.; Yan, Y. F.; Xiong, R. G. An organic-inorganic perovskite ferroelectric with large piezoelectric response. *Science* **2017**, *357*, 306−309.

(8) Mitzi, D. B.; Field, C. A.; Harrison, W. T.; Guloy, A. M. Conducting tin halides with a layered organic-based perovskite structure. *Nature* **1994**, *369*, 467−469.

(9) Mitzi, D. B.; Wang, S.; Field, C. A.; Chess, C. A.; Guloy, A. M. Conducting Layered Organic-inorganic Halides Containing < 110>-Oriented Perovskite Sheets. *Science* **1995**, *267*, 1473−1476.

(10) Takahashi, Y.; Obara, R.; Nakagawa, K.; Nakano, M.; Tokita, J. Y.; Inabe, T. Tunable Charge Transport in Soluble Organic−Inorganic Hybrid Semiconductors. *Chem. Mater.* **2007**, *19*, 6312−6316.

(11) Scaife, D. E.; Weller, P. F.; Fisher, W. G. Crystal preparation and properties of cesium tin(II) trihalides. *J. Solid State Chem.* **1974**, *9*, 308−314.

(12) Chung, I.; Song, J. H.; Im, J.; Androulakis, J.; Malliakas, C. D.; Li, H.; Freeman, A. J.; Kenney, J. T.; Kanatzidis, M. G. $CsSnI_3$: Semiconductor or Metal? High Electrical Conductivity and Strong Near-Infrared Photoluminescence from a Single Material. High Hole Mobility and Phase-Transitions. *J. Am. Chem. Soc.* **2012**, *134*, 8579−8587.

(13) Kinbara, K.; Aida, T. Toward Intelligent Molecular Machines: Directed Motions of Biological and Artificial Molecules and Assemblies. *Chem. Rev.* **2005**, *105*, 1377−1400.

(14) Dominguez, Z.; Khuong, T. A.; Dang, H.; Sanrame, C. N.; Nuñez, J. E.; Garcia-Garibay, M. A. Molecular Compasses and Gyroscopes with Polar Rotors: Synthesis and Characterization of Crystalline Forms. *J. Am. Chem. Soc.* **2003**, *125*, 8827−8837.

(15) Leigh, D. A.; Wong, J. K.; Dehez, F.; Zerbetto, F. Unidirectional rotation in a mechanically interlocked molecular rotor. *Nature* **2003**, *424*, 174−179.

(16) Akutagawa, T.; Koshinaka, H.; Sato, D.; Takeda, S.; Noro, S. I.; Takahashi, H.; Kumai, R.; Tokura, Y.; Nakamura, T. Ferroelectricity and polarity control in solid-state flip-flop supramolecular rotators. *Nat. Mater.* **2009**, *8*, 342−347.

(17) Frost, J. M.; Butler, K. T.; Brivio, F.; Hendon, C. H.; van Schilfgaarde, M.; Walsh, A. Atomistic Origins of High-Performance in Hybrid Halide Perovskite Solar Cells. *Nano Lett.* **2014**, *14*, 2584−2590.

(18) Juarez-Perez, E. J.; Sanchez, R. S.; Badia, L.; Garcia-Belmonte, G.; Kang, Y. S.; Moera-Sero, I.; Bisquert, J. Photoinduced Giant Dielectric Constant in Lead Halide Perovskite Solar Cells. *J. Phys. Chem. Lett.* **2014**, *5*, 2390−2394.

(19) Sato, O. Dynamic molecular crystals with switchable physical properties. *Nat. Chem.* **2016**, *8*, 644−656.

(20) Zhang, W.; Xiong, R. G. Ferroelectric Metal−Organic Frameworks. *Chem. Rev.* **2012**, *112*, 1163−1195.

(21) Zhang, Y.; Liao, W. Q.; Fu, D. W.; Ye, H. Y.; Chen, Z. N.; Xiong, R. G. Highly Efficient Red-Light Emission in An Organic−Inorganic Hybrid Ferroelectric: (Pyrrolidinium)$MnCl_3$. *J. Am. Chem. Soc.* **2015**, *137*, 4928−4931.

(22) Sun, Z. H.; Chen, T. L.; Luo, J. H.; Hong, M. C. Bis(imidazolium) L-Tartrate: A Hydrogen-Bonded Displacive-Type Molecular Ferroelectric Material. *Angew. Chem., Int. Ed.* **2012**, *51*, 3871−3876.

(23) Sun, Z. H.; Luo, J. H.; Chen, T. L.; Li, L. N.; Xiong, R. G.; Tong, M. L.; Hong, M. C. Distinct Molecular Motions in a Switchable Chromophore Dielectric 4-N,N-Dimethylamino-4-N-methylstilbazolium Trifluoromethanesulfonate. *Adv. Funct. Mater.* **2012**, *22*, 4855−4861.

(24) Sun, Z. H.; Tang, Y. Y.; Zhang, S. Q.; Ji, C. M.; Chen, T. L.; Luo, J. H. Ultrahigh Pyroelectric Figures of Merit Associated with Distinct Bistable Dielectric Phase Transition in a New Molecular Compound: Di-n-Butylaminium Trifluoroacetate. *Adv. Mater.* **2015**, *27*, 4795−4801.

(25) Liao, W. Q.; Zhang, Y.; Hu, C. L.; Mao, J. G.; Ye, H. Y.; Li, P. F.; Huang, S. D.; Xiong, R. G. A lead-halide perovskite molecular ferroelectric semiconductor. *Nat. Commun.* **2015**, *6*, 7338.

DOI: 10.1021/acs.inorgchem.7b01863
Inorg. Chem. 2017, 56, 13078−13085

(26) Ye, H. Y.; Liao, W. Q.; Hu, C. L.; Zhang, Y.; You, Y. M.; Mao, J.G.; Li, P. F.; Xiong, R. G. Bandgap Engineering of Lead-Halide Perovskite-Type Ferroelectrics. *Adv. Mater.* **2016**, *28*, 2579−2586.
(27) Lee, M. M.; Teuscher, J.; Miyasaka, T.; Murakami, T. N.; Snaith, H. J. Efficient Hybrid Solar Cells Based on Meso-Superstructured Organometal Halide Perovskites. *Science* **2012**, *338*, 643−647.
(28) Stranks, S. D.; Eperon, G. E.; Grancini, G.; Menelaou, C.; Alcocer, M. J.; Leijtens, T.; Herz, L. M.; Petrozza, A.; Snaith, H. J. Electron-Hole Diffusion Lengths Exceeding **1** Micrometer in an Organometal Trihalide Perovskite Absorber. *Science* **2013**, *342*, 341−344.
(29) Yang, W. S.; Noh, J. H.; Jeon, N. J.; Kim, Y. C.; Ryu, S.; Seo, J.; Seok, S. I. High-performance photovoltaic perovskite layers fabricated through intramolecular exchange. *Science* **2015**, *348*, 1234−1237.
(30) Jeon, N. J.; Noh, J. H.; Yang, W. S.; Kim, Y. C.; Ryu, S.; Seo, J.; Seok, S. I. Compositional engineering of perovskite materials for high-performance solar cells. *Nature* **2015**, *517*, 476−480.
(31) Etgar, L.; Gao, P.; Xue, Z. S.; Peng, Q.; Chandiran, A. K.; Liu, B.; Nazeeruddin, M. K.; Grätzel, M. Mesoscopic $CH_3NH_3PbI_3/TiO_2$ Heterojunction Solar Cells. *J. Am. Chem. Soc.* **2012**, *134*, 17396−17399.
(32) Hsu, H. Y.; Ji, L.; Ahn, H. S.; Zhao, J.; Yu, E. T.; Bard, A. J. A Liquid Junction Photoelectrochemical Solar Cell Based on p-Type $MeNH_3PbI_3$ Perovskite with 1.05 V Open-Circuit Photovoltage. *J. Am. Chem. Soc.* **2015**, *137*, 14758−14764.
(33) Kojima, A.; Teshima, K.; Shirai, Y.; Miyasaka, T. Organometal Halide Perovskites as Visible-Light Sensitizers for Photovoltaic Cells. *J. Am. Chem. Soc.* **2009**, *131*, 6050−6051.
(34) Cho, H.; Jeong, S. H.; Park, M. H.; Kim, Y. H.; Wolf, C.; Lee, C. L.; Heo, G. H.; Sadhanala, A.; Myoung, N.; Yoo, S.; Im, S. H.; Friend, R. H.; Lee, T. W. Overcoming the electroluminescence efficiency limitations of perovskite light-emitting diodes. *Science* **2015**, *350*, 1222−1225.
(35) Pathak, S.; Sakai, N.; Wisnivesky Rocca Rivarola, F.; Stranks, S. D.; Liu, J.; Eperon, G. E.; Ducati, C.; Wojciechowski, K.; Griffiths, J. T.; Haghighirad, A. A.; Pellaroque, A.; Friend, R. H.; Snaith, H. J. Perovskite Crystals for Tunable White Light Emission. *Chem. Mater.* **2015**, *27*, 8066−8075.
(36) Deschler, F.; Price, M.; Pathak, S.; Klintberg, L. E.; Jarausch, D.-D.; Higler, R.; Hüttner, S.; Leijtens, T.; Stranks, S. D.; Snaith, H. J.; Atatüre, M.; Phillips, R. T.; Friend, R. H. High Photoluminescence Efficiency and Optically Pumped Lasing in Solution-Processed Mixed Halide Perovskite Semiconductors. *J. Phys. Chem. Lett.* **2014**, *5*, 1421−1426.
(37) Sutherland, B. R.; Hoogland, S.; Adachi, M. M.; Wong, C. T.; Sargent, E. H. Conformal Organohalide Perovskites Enable Lasing on Spherical Resonators. *ACS Nano* **2014**, *8*, 10947−10952.
(38) Deretzis, L.; La Magna, A. L. Exploring the orthorhombic−tetragonal phase transition in $CH_3NH_3PbI_3$: the role of atom kinetics. *Nanoscale* **2017**, *9*, 5896−5903.
(39) Chen, T.; Foley, B. J.; Ipek, B.; Tyagi, M.; Copley, J. R.; Brown, C. M.; Choi, J. J.; Lee, S. H. Rotational dynamics of organic cations in the $CH_3NH_3PbI_3$ perovskite. *Phys. Chem. Chem. Phys.* **2015**, *17*, 31278−31286.
(40) Guo, Z.; Yoon, S. J.; Manser, J. S.; Kamat, P. V.; Luo, T. Structural Phase- and Degradation-Dependent Thermal Conductivity of $CH_3NH_3PbI_3$ Perovskite Thin Films. *J. Phys. Chem. C* **2016**, *120*, 6394−6401.
(41) Stoumpos, C. C.; Malliakas, C. D.; Kanatzidis, M. G. Semiconducting Tin and Lead Iodide Perovskites with Organic Cations: Phase Transitions, High Mobilities, and Near-Infrared Photoluminescent Properties. *Inorg. Chem.* **2013**, *52*, 9019−9038.
(42) Stroppa, A.; Quarti, C.; De Angelis, F.; Picozzi, S. Ferroelectric Polarization of $CH_3NH_3PbI_3$: A Detailed Study Based on Density Functional Theory and Symmetry Mode Analysis. *J. Phys. Chem. Lett.* **2015**, *6*, 2223−2231.
(43) Röhm, H.; Leonhard, T.; Hoffmann, M. J.; Colsmann, A. Ferroelectric domains in methylammonium lead iodide perovskite thin-films. *Energy Environ. Sci.* **2017**, *10*, 950−955.
(44) Fan, Z.; Xiao, J.; Sun, K.; Chen, L.; Hu, Y.; Ouyang, J.; Ong, K. P.; Zeng, K.; Wang, J. Ferroelectricity of $CH_3NH_3PbI_3$ Perovskite. *J. Phys. Chem. Lett.* **2015**, *6*, 1155−1161.
(45) Meng, X.; Zhang, R. F.; Fu, Z. H.; Zhang, Q. F. Domain-dependent electronic structure and optical absorption property in hybrid organic−inorganic perovskite. *Phys. Chem. Chem. Phys.* **2016**, *18*, 27358−27365.
(46) Rothmann, M. U.; Li, W.; Zhu, Y.; Bach, U.; Spiccia, L.; Etheridge, J.; Cheng, Y. B. Direct observation of intrinsic twin domains in tetragonal $CH_3NH_3PbI_3$. *Nat. Commun.* **2017**, *8*, 14547.
(47) Kutes, Y.; Ye, L.; Zhou, Y.; Pang, S.; Huey, B. D.; Padture, N. P. Direct Observation of Ferroelectric Domains in Solution-Processed $CH_3NH_3PbI_3$ Perovskite Thin Films. *J. Phys. Chem. Lett.* **2014**, *5*, 3335−3339.
(48) Stroppa, A.; Di Sante, D.; Barone, P.; Bokdam, M.; Kresse, G.; Franchini, C.; Whangbo, M. H.; Picozzi, S. Tunable ferroelectric polarization and its interplay with spin−orbit coupling in tin iodide perovskites. *Nat. Commun.* **2014**, *5*, 5900.
(49) Harada, J.; Shimojo, T.; Oyamaguchi, H.; Hasegawa, H.; Takahashi, Y.; Satomi, K.; Suzuki, Y.; Kawamata, J.; Inabe, T. Directionally tunable and mechanically deformable ferroelectric crystals from rotating polar globular ionic molecules. *Nat. Chem.* **2016**, *8*, 946−952.
(50) Bennett, T. D.; Cheetham, A. K.; Fuchs, A. H.; Coudert, F. X. Interplay between defects, disorder and flexibility in metal-organic frameworks. *Nat. Chem.* **2016**, *9*, 11−16.
(51) Yin, Y. W.; Burton, J. D.; Kim, Y. M.; Borisevich, A. Y.; Pennycook, S. J.; Yang, S. M.; Noh, T. W.; Gruverman, A.; Li, X. G.; Tsymbal, E. Y.; Li, Q. Enhanced tunnelling electroresistance effect due to a ferroelectrically induced phase transition at a magnetic complex oxide interface. *Nat. Mater.* **2013**, *12*, 397−402.
(52) Zhang, W. Y.; Tang, Y. Y.; Li, P. F.; Shi, P. P.; Liao, W. Q.; Fu, D. W.; Ye, H. Y.; Zhang, Y.; Xiong, R. G. Precise Molecular Design of High-*Tc* 3D Organic-Inorganic Perovskite Ferroelectric: [MeHdabco]-RbI_3 (MeHdabco = *N*-Methyl-1,4-diazoniabicyclo[2.2.2]octane). *J. Am. Chem. Soc.* **2017**, *139*, 10897−10902.
(53) Pan, Q.; Liu, Z. B.; Tang, Y. Y.; Li, P. F.; Ma, R. W.; Wei, R. Y.; Zhang, Y.; You, Y. M.; Ye, H. Y.; Xiong, R. G. A Three-Dimensional Molecular Perovskite Ferroelectric: (3-Ammoniopyrrolidinium)$RbBr_3$. *J. Am. Chem. Soc.* **2017**, *139*, 3954−957.
(54) Xu, W. J.; Li, P. F.; Tang, Y. Y.; Zhang, W. X.; Xiong, R. G.; Chen, X. M. A Molecular Perovskite with Switchable Coordination Bonds for High-Temperature Multiaxial Ferroelectrics. *J. Am. Chem. Soc.* **2017**, *139*, 6369−6375.
(55) Shi, P. P.; Tang, Y. Y.; Li, P. F.; Liao, W. Q.; Wang, Z. X.; Ye, Q.; Xiong, R. G. Symmetry breaking in molecular ferroelectrics. *Chem. Soc. Rev.* **2016**, *45*, 3811−3827.
(56) Sun, Z. H.; Zeb, A.; Liu, S. J.; Ji, C. M.; Khan, T.; Li, L. N.; Hong, M. C.; Luo, J. H. Exploring a Lead-free Semiconducting Hybrid Ferroelectric with a Zero-Dimensional Perovskite-like Structure. *Angew. Chem., Int. Ed.* **2016**, *55*, 11854−11858.
(57) Ji, C.; Sun, Z.; Zeb, A.; Liu, S.; Zhang, J.; Hong, M.; Luo, J. Bandgap Narrowing of Lead-Free Perovskite-Type Hybrids for Visible-Light-Absorbing Ferroelectric Semiconductors. *J. Phys. Chem. Lett.* **2017**, *8*, 2012−2018.
(58) Zhang, X.; Shao, X. D.; Li, S. C.; Cai, Y.; Yao, Y. F.; Xiong, R. G.; Zhang, W. Dynamics of a caged imidazolium cation−toward understanding the order-disorder phase transition and the switchable dielectric constant. *Chem. Commun.* **2015**, *51*, 4568−4571.
(59) Li, P.; Yang, X.; Maβ, T. W. W.; Hanss, J.; Lewin, M.; Michel, A. K. U.; Wuttig, M.; Taubner, T. Reversible optical switching of highly confined phonon−polaritons with an ultrathin phase-change material. *Nat. Mater.* **2016**, *15*, 870−875.
(60) Raeliarijaona, A.; Fu, H. Persistence of strong and switchable ferroelectricity despite vacancies. *Sci. Rep.* **2017**, *7*, 41301.
(61) Itkis, M. E.; Chi, X.; Cordes, A. W.; Haddon, R. C. Magneto-Opto-Electronic Bistability in a Phenalenyl-Based Neutral Radical. *Science* **2002**, *296*, 1443−1445.

DOI: 10.1021/acs.inorgchem.7b01863
Inorg. Chem. 2017, 56, 13078−13085

(62) Kahn, O.; Martinez, C. J. Spin-Transition Polymers: From Molecular Materials Toward Memory Devices. *Science* **1998**, *279*, 44−48.
(63) Bibes, M.; Barthélémy, A. Multiferroics: Towards a magnetoelectric memory. *Nat. Mater.* **2008**, *7*, 425−426.
(64) Sheldrick, G. M. *SHELXL-97: Programs for X-ray Crystal*; Bruker AXS: Madison, WI, 1997.
(65) Wang, D. Y.; Qin, L. L.; Hou, X. L.; Ye, H. Y. Temperature-triggered phase transition in pyridazine hexafluorophosphate. *Chin. Chem. Lett.* **2015**, *26*, 145−148.
(66) Wang, H. T.; Kong, L. H.; Shi, P. P.; Li, Q.; Ye, Q.; Fu, D. W. The structure and dielectric properties of ionic compounds with flexible ammonium moiety. *Chin. Chem. Lett.* **2015**, *26*, 382−386.
(67) Wang, F. F.; Chen, C.; Zhang, Y.; Fu, D. W. Crystal structure and dielectric property of supramolecular macrocyclic $[(NDPA)\cdot(18\text{-crown-}6)]^{2+}\cdot(DMA)^{+}\cdot 3ClO_4^{-}$ assemblies. *Chin. Chem. Lett.* **2015**, *26*, 31−35.
(68) Zhang, W.; Cai, Y.; Xiong, R. G.; Yoshikawa, H.; Awaga, K. Exceptional Dielectric Phase Transitions in a Perovskite-Type Cage Compound. *Angew. Chem., Int. Ed.* **2010**, *49*, 6608−6610.
(69) Lin, L.; Yu, R.; Yang, W.; Wu, X. Y.; Lu, C. Z. A Series of Chiral Metal−Organic Frameworks Based on Oxalyl Retro-Peptides: Synthesis, Characterization, Dichroism Spectra, and Gas Adsorption. *Cryst. Growth Des.* **2012**, *12*, 3304−3311.
(70) Govorov, A. O.; Fan, Z.; Hernandez, P.; Slocik, J. M.; Naik, R. R. Theory of Circular Dichroism of Nanomaterials Comprising Chiral Molecules and Nanocrystals: Plasmon Enhancement, Dipole Interactions, and Dielectric Effects. *Nano Lett.* **2010**, *10*, 1374−1382.
(71) Fröhlich, H. *Theory of dielectrics*, 2nd ed.; Oxford University Press: Oxford, U.K., 1965.
(72) Tauc, J.; Grigorovici, R.; Vancu, A. Optical Properties and Electronic Structure of Amorphous Germanium. *Phys. Status Solidi B* **1966**, *15*, 627−637.
(73) Kortüm, G.; Braun, W.; Herzog, G. Prinzip und Meßmethodik der diffusen Reflexionsspektroskopie. *Angew. Chem.* **1963**, *75*, 653−661.
(74) Liu, Y. C.; Yang, Z.; Cui, D.; Ren, X. D.; Sun, J. K.; Liu, X. J.; Zhang, J. R.; Wei, Q. B.; Fan, H. B.; Yu, F. Y.; Zhang, X.; Zhao, C. M.; Liu, S. D. Two-Inch-Sized Perovskite $CH_3NH_3PbX_3$ (X = Cl, Br, I) Crystals: Growth and Characterization. *Adv. Mater.* **2015**, *27*, 5176−5183.
(75) Kuang, W.; Nelson, S. O. Low-frequency dielectric properties of biological tissues: a review with some new insights. *Trans. ASAE* **1998**, *41*, 173−184.
(76) Kremer, F.; Schonhals, A. *Broadband Dielectric Spectroscopy*; Springer: Berlin, 2002.
(77) Yuan, Y. B.; Li, T.; Wang, Q.; Xing, J.; Gruverman, A.; Huang, J. S. Anomalous photovoltaic effect in organic-inorganic hybrid perovskite solar cells. *Sci. Adv.* **2017**, *3*, e1602164.
(78) Xu, Y. X.; Ye, Y.; Liu, T. F.; Wang, X. L.; Zhang, B. Q.; Wang, M.; Han, H. X.; Li, C. Unraveling a Single-Step Simultaneous Two-Electron Transfer Process from Semiconductor to Molecular Catalyst in a CoPy/CdS Hybrid System for Photocatalytic H_2 Evolution under Strong Alkaline Conditions. *J. Am. Chem. Soc.* **2016**, *138*, 10726−10729.
(79) Yin, W. J.; Yang, J. H.; Kang, J.; Yan, Y.; Wei, S. H. Halide perovskite materials for solar cells: a theoretical review. *J. Mater. Chem. A* **2015**, *3*, 8926−8942.

DOI: 10.1021/acs.inorgchem.7b01863
Inorg. Chem. 2017, 56, 13078−13085